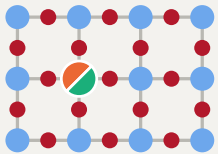
Pr and Rb lower the O-vacancy formation energy; only Rb makes it negative

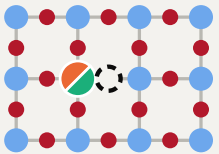
Anatase TiO_2 · PBE+U · neutral V_O · O-rich limit $\mu_O = E(O_2)/2$

❶ Doping



Ti → Pr^{3+} or Rb^{+} (48 atoms)

❷ O vacancy



Neutral V_O donates ●● 2 e^-

❸ Method

U(Ti-3d) 3.5 eV
Γ-point, 40 Ry

✓ k-mesh, cut-off, D3: Pr–Rb Δ shifts ≤ 0.04 eV

! Smearing: Pr cells polarise; E_f(Pr) +0.99 eV

❹ Charge balance

○ hole ● electron
formal counts

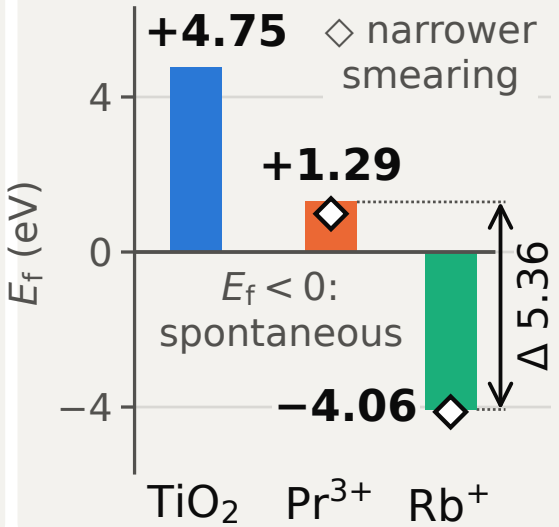
● Pr^{3+}

○ + ●● → ●

net: 1 e^- (Ti-3d)

● Rb^{+}

○○○ + ●● → ○

net: 1 h^+ (O-2p)

❺ Formation energy